## Quotation No : Q24000148

Date : 20-Mar-2024

MAW Engineering Pvt Ltd
Kathmandu
MAYAGDI
NEPAL

## Kind Attention : Mr./Ms. Suraj Puri

Dear Sir,

We understand that in industry, reliability is not just a requirement,it is the foundation of success.
That is why Absstem Industrial PSA Generator is more than just a product; it is a commitment to excellence, efficiency, and uninterrupted performance.

Every component is designed with precision and innovation, ensuring that each molecule of oxygen and nitrogen is delivered with the utmost purity.

With fully automated operation, our system seamlessly integrates into your processes, reducing manual intervention and maximizing productivity.

Moreover, our industry-leading air ratio ensures that you achieve the highest efficiency, optimizing energy consumption while maintaining peak performance.
And because we believe in standing by our craftsmanship, we offer an unmatched-10 year warranty on sieves, giving you the peace of mind that your investment is protected for the long run.

We have attached the detailed quotation for your review.

Should you have any questions or require further customization to meet your specific needs, please do not hesitate to reach out.

We are here to provide not just a product but a solution that breathes life into your operations.

We look forward to the opportunity to serve you and build a lasting partnership.

Thanks and Regards,

## For Absstem Industrial Solutions Pvt. Ltd.

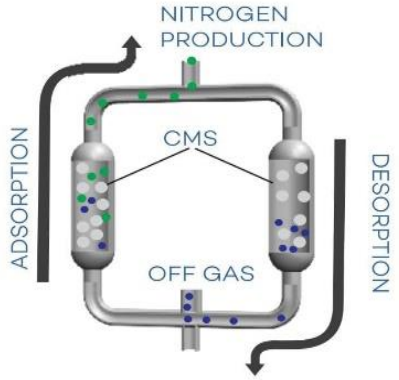
Arifa Zaidi
Assistant Manager Business Development
9218108184
www.absstem.com

nGENX POWERED BY ABSSTEM INDUSTRIAL

# Industrial
# PSA Nitrogen Generator




## PSA Technology




PSA (PRESSURE SWING ADSORPTION)

Industrial Nitrogen Generator mainly consists of two adsorbers filled with carbon molecular sieves. Pressurized air is passed through one of these adsorbers, carbon having the property to attract oxygen, adsorbs oxygen from air and Nitrogen is allowed to pass through and move into the storage tank as the final product. As the adsorber approaches saturation, the air is allowed to pass through the other adsorber while this adsorber is being regenerated to ensure a continuous supply of Nitrogen. This changeover takes place through a specially designed microcontroller to ensure a hassle-free operation. The quality of Nitrogen produced is monitored by a Nitrogen analyser.

- Shipped ready for installation
- 5 minutes to run and never have to stop it!
- Easy installation
- Low operating expenses compared to gas cylinders/liquid Nitrogen
- Maintenance costs are very less
- Embedded Safety Interlocks
- Space-saving design
- Eliminates all hazards which are there with gas cylinder or liquid Nitrogen
- Longevity: at least 10 years

## Description

| | | |
|---|---|---|
| a. | Product Name | Nitrogen Gas Generator |



**Absstem Industrial Solutions Pvt. Ltd.**
Plot No. 591, Sector 8, Phase IV
IMT Manesar, Gurugram, Haryana - 122051
Contact - 1800 3010 3394 Email - indsales@absstem.com
CIN: U29256HR2022PTC103387

b. Model Range nGENX
Technology PSA (Pressure Swing Adsorption)

## Specifications

| | | |
|---|---|---|
| **a.** | Nitrogen Flow Rate | As per Performance Sheet |
| **b.** | Nitrogen Purity | As per Performance Sheet |
| **c.** | Nitrogen Delivery Pressure | 6.5 Barg (Higher Pressure Options Available) |
| **d.** | Ambient conditions | |
| | Temperature | 0 to 50 Deg. Celsius |
| | Relative Humidity | up to 80 % (Max.) |
| | Altitude | up to 1000 m above Sea level |

## Scheme Offered – (PSA Only)

This scheme will comprise of

- Air package with inlet air filtration system
  (Air compressor, Air dryer, Water Separator, Oil filters)
- Industrial Nitrogen generator
- Storage Tank
- Nitrogen Analyzer
- Alarm panel

## Unique Features of Absstem

a. All the fabricated items are cleaned and degreased following our strict quality policy.
b. All the Instruments are specially manufactured to make them suitable for Nitrogen gas.
c. Guaranteed 10+ years of healthy life.
d. We provide the purest **Nitrogen** in Industry.
e. Data logging and Graphical Reports for all parameters available.
f. Test certificates of all instruments are available on demand.

## **Onsite Gas Generation** vs Liquid Tank/Cylinder Alternative

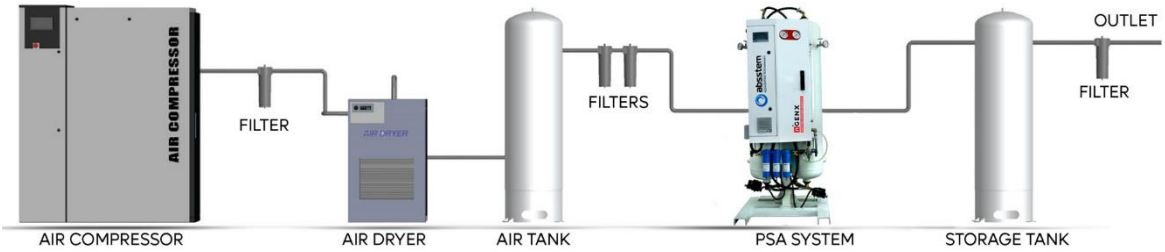
| |
|---|
| *Non-stop availability: 24 hours a day, 7 days a week* |
| *Easy integration within existing installations* |
| *Your own independent supply of industrial gas* |
| *Significant economies for scale and lower operational costs: no rental charges, logistical expenses and bulk user evaporation losses* |
| *No safety hazards when handling high-pressure cylinders* |



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## Equipment Description

The Industrial Nitrogen Generator consists of




| | |
|---|---|
| Air Compressor | An Air compressor is used to compress the atmospheric air to a pressure required for the separation of **Nitrogen** from the air. |
| Refrigeration dryer | Compressed air then passes through a refrigerated dryer to remove moisture (which is harmful to sieves) from the air. |
| Air Tank | Dry air is collected in an air tank to remove the pulsations which serve the dynamic (changing) requirement of air by the PSA system. |
| Multistage Filtration system | Before air enters the PSA system, it is passed through multistage filtration removing moisture, particulates and oil content to the specified levels. |
| PSA system | Pressurized air is passed through PSA Towers filled with Molecular sieves to separate **Nitrogen** from Air. |
| Storage Tank | Industrial Grade **Nitrogen** produced from NGEN is stored in this tank to supply a continuous flow of **Nitrogen**. |
| Nitrogen Analyzer | Nitrogen produced is monitored by a **Nitrogen** analyser. In case, O2 purity goes below the specified level, it gives an alarm and switches it to vent mode. |

## EYE-SMART Control System

(User-friendly interface for easy operation and maintenance)

Absstem industrial Nitrogen generator complies with all industry standards and regulation

- Alarm management system
- Plug and play
- Fully automatic
- Process data display
- Consumables running hours display
- Data logging




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## Additional upgrades

- Digital flow meter with totaliser
- High-pressure Nitrogen tank
- Cylinder manifold and alarm management system
- Remote monitoring- IoT Ready

## Plant Operation

- Unit operates 24-hrs a day, 7-days a week reliably, automatically & safely

## Automation

- The Plant is fully automatic and needs only electrical power for running the Air compressor
- The machine automatically enters into standby mode when the level in the Storage Tank reaches its set limit. This helps to increase the life of the Air compressor.

## International Quality standards

A strict quality procedure, following the leading international pharmacopoeia and Industrial directives, is followed in the selection of all mechanical and electrical instruments, vessel fabrication and designing of Industrial Nitrogen generators.

## Itemised Supply description

| No. | Item description | Make | Qty |
|---|---|---|---|
| a | Air Compressor with Filters | ABSSTEM APPROVED | Client Scope / Quoted Separately |
| b | Refrigeration Dryer | ABSSTEM APPROVED | Client Scope / Quoted Separately |
| c | Air Tank | ABSSTEM | Client Scope / Quoted Separately |
| d | Set of Filters | ABSSTEM | 1 no. |
| e | PSA unit (NGEN) | ABSSTEM | 1 no. |
| f | Electrical Panel | ABSSTEM | 1 no. |
| f | Digital Nitrogen Analyzer | ABSSTEM | 1 no. |
| g | Storage Tank | ABSSTEM | 1 no. |
| h | Outlet product gas Filter | ABSSTEM | 1 no. |



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## Electrical Protection

The client should make proper arrangements for earthing/grounding systems for the Industrial **Nitrogen** Generator at their facility.

## Enclosure Panel

- Ergonomically designed metal enclosure with powder-coated finish for longer life.
- High-strength IP-rated panel fittings from Germany for a robust enclosure.
- Fan filters IP-55 rated making the electrical enclosure's internal environment dust-free.
- Easy connectivity to any place of installation with req. port connections.
- High-strength base frame designed for higher load and easy transportation.
- Vessels are designed per ASME section VIII DIV I. Vessels are epoxy paint coated which is considered good for longevity and corrosion resistance.
- All electrical items are CE certified with operating voltage at 24 VDC

## Annual Maintenance Contract (AMC)

Our in-house support Absstem shield is 24*7 open and we assure a timely service maintenance schedule. Per the schedule mentioned in the operational manual, we suggest the operator change the filters by following a simple protocol.

It is very easy to change or replace as our commissioning engineer plus our commissioning engineer and tech support office will always guide you about the activity.

## Best ROI Guaranteed

*ROI within 6 to 18 months when compared to conventional methods*

Generator performance depends on the component selection. Our processes make this job easier as all equipment is selected based on more than 3 decades of research. NGEN is a fully automatic (operator-free operation) that ensures peace of mind.

# We deliver what we commit!!

Thank You



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# MINIMUM SPECIFICATIONS FOR FOOD GAS APPLICATIONS

## Doc 126/18

Revision of Doc 126/11

***EUROPEAN INDUSTRIAL GASES ASSOCIATION AISBL***

AVENUE DES ARTS 3-5 • B – 1210 BRUSSELS
Tel: +32 2 217 70 98 • Fax: +32 2 219 85 14
E-mail: info@eiga.eu • Internet: www.eiga.eu

## Appendix 1 - Summary of Current Gases Specifications in EU Legislation and JECFA

| Component / Impurity | Standard | Carbon dioxide E 290 | Nitrogen E 941 | Oxygen E 948 | Argon E 938 | Helium E 939 | Nitrous oxide E 942 | n-/iso-Butane E 943 a/b | Propane E 944 | Hydrogen E 949 | Sulfur dioxide E 220 |
|---|---|---|---|---|---|---|---|---|---|---|---|
| Assay (v/v) | EC | >99% | >99% | >99% | >99% | >99% | >99% | >96%/>94% | >95% | >99,9% | >99 % |
| | JECFA | >99% | >99% | >99% | >99% | >99% | >97% | | | | >99,9 % |
| Odour | EC | | | | | | | | | | |
| | JECFA | | | Free | free | free | | | | | |
| Moisture | EC | | <0,05% | <0,05% | <0,05% | <0,05% | <0,05% | <50vppm | <50vppm | <50vppm | <0,05% |
| | JECFA | <52 vppm | | | | | | | | | <0,05% |
| $CO_2$ | EC | | | | | | | | | | |
| | JECFA | | | <300 vppm | | | | | | | |
| CO | EC | <10 vppm | <10 vppm | | | | <30 vppm | | | | |
| | JECFA | <10 vppm | <10 vppm | <10 vppm | | <10 vppm | <10 vppm | | | | |
| $NO/NO_2$ | EC | | <10 vppm | | | | <10 vppm | | | | |
| | JECFA | | | | | | <5 vppm | | | | |
| Total Hydrocarbon | EC | | <100 vppm | <100 vppm | <100 vppm | <100 vppm | | see * | see ** | | |
| | JECFA | <50vppm | | | | | | | | | |
| Residual Gases ($O_2$, $N_2$, $H_2$) | EC | | < 1 % (O2) | | | | | | | <0.07% | |
| | JECFA | | | | <1% | | | | | | |
| Oil | EC | <5mg/kg | | | | | | | | | |
| | JECFA | <10 ppmw | | | | | | | | | |
| Acidity & Red. Subst. | EC | pass test | | | | | | | | | |
| | JECFA | pass test | | | | | | | | | |
| Halogens & $H_2S$ | EC | | | | | | | | | | |
| | JECFA | | | | | | <5 vppm | | | | |
| Other comp. & Heavy metals | EC | | | | | | | | | | see*** |
| | JECFA | | | | | | | | | | see**** |

* Methane < 0,15%, other hydrocarbons < 5,1%. For E943a the maximum limit of hydrocarbons are:
$C_2H_6$ < 0,5%;
$C_3H_8$<1,5%;
i-$C_4H_{10}$<3%,
1.3-$C_4H_6$<0,1%

** Methane < 0,15%, other hydrocarbons are < 6,6%. For E943b the maximum limit of hydrocarbons are:
$C_2H_6$ < 0,5%;
$C_3H_8$<2,0%;
i-$C_4H_{10}$<4%,
1.3-$C_4H_6$<0,1%

*** EC Directive 2008/84 [2]: Non-volatile residue <0.001 %, Sulphur trioxide < 0,1 %, Selenium < 10 mg/kg, Arsenic <3mg/kg, Lead <5mg/kg, Mercury <1mg/kg, Heavy metals (as Pb) <10 mg/kg

**** JECFA (1998) [3]: Non-volatile residue <0,05%, Selenium <20 mg/kg, Lead <5mg/kg ,Other gases not present in air : No trace

# Product Description

| Product | Description | Make | Qty |
|---|---|---|---|
| nGenx Nitrogen Generator | nGenx 05<br>Nitrogen | | 1 |

| Caption | Specification |
|---|---|
| Gas Generated | Nitrogen |
| Gas Pressure | upto 6.5 Barg( Higher options Available) |
| Flow @ 95 % | 25 Nm3/Hr |
| Flow @ 99 % | 20 Nm3/Hr |
| Flow @ 99.5 % | 17 Nm3/Hr |
| Flow @ 99.9 % | 15 Nm3/Hr |
| Compressed Air Requirement | 37 CFM |
| Module | PSA Plant |
| Air Compressor And Drier | In your Scope / Quoted Separately |

| Product | Description | Make | Qty |
|---|---|---|---|
| nGen Gas Storage | nGEN 01 / 02 / 04 ST<br>Nitrogen | | 1 |

| Caption | Specification |
|---|---|
| Water Volume | 500 Liters |
| Gas Stored | Nitrogen |
| Gas Stored Volume | 2.75 M3 |
| Working Pressure | 6.5 Barg |
| Hydrotest Pressure | 9 Barg |



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| Product | Description | Make | Qty |
|---|---|---|---|
| nGen Air Package | nGEN 07 CP / nGenx 05 CP Nitrogen | | 1 |

| Caption | Specification |
|---|---|
| Air Compressor KW | 7.5 KW |
| Air Compressor Make | Atlas Copco/ Ingersoll Rand/ Gardner Denver/ As per Demand |
| Air Dryer | Included |
| Dry Air Storage | Included |
| Interconnecting Piping | Included |
| Pre filter | Included |
| Dry Air Generated | 43 CFM |

## Commercial Offer

| S No | Item Code | Item Name | Description | HSN | Qty | Unit Price | Discount | GST(%) | Amount |
|---|---|---|---|---|---|---|---|---|---|
| 1 | nGenx 05 | nGenx 05 | nGenx Nitrogen Generator | 840510 | 1 | 7,28,400.00 | 0.00 | 18.00 | 7,28,400.00 |
| 2 | nGEN 01 ST | nGEN 01 / 02 / 04 ST | nGen Gas Storage | 840510 | 1 | 60,000.00 | 0.00 | 18.00 | 60,000.00 |
| 3 | nGEN 07 CP | nGEN 07 CP / nGenx 05 CP | nGen Air Package | 840510 | 1 | 6,15,000.00 | 2,10,000.00 | 18.00 | 4,05,000.00 |
| | | | | | | | | Sub Total | 11,93,400.00 |
| | | | | | | | | GST(%) @18.00 | 2,14,812.00 |
| | | | | | | | | Grand Total () | 14,08,212.00 |



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# Terms And Conditions

| | | |
|---|---|---|
| 1. | EXCLUSIONS & CLIENTS SCOPE | All erection / civil work, All Power Cables & Cabling out of battery limit of PSA Plant, Cable Trenches etc. Unloading / Shifting of Gas Plant at Clients site will be in Client Scope. Additional gas piping, cooling water (if required). |
| 2. | SUPERVISION OF ERECTION & COMMISSIONING | There shall be no commissioning charges, however you shall arrange freeTo & Fro Air Ticket, lodging / boarding, and local conveyance for our Engineers during supervision of Installation of Gas Plant at site on free of cost basis. Our Engineers will be there for Supervision only. Any manpower / fitter required for installation work shall be provided by you. |
| 3. | P&F | 3% Extra. |
| 4. | Freight | Extra at Actuals. |
| 5. | Insurance | In your Scope. |
| 6. | Payment Terms | 50% Advance with your Prestegious Purchase Order and Balance against PI. |
| 7. | Delivery | 6-8 Weeks from the PO and Advance. Rest can be confirmed for Availability. |
| 8. | Warranty | We hereby confirm that, our equipment shall have Suppliers warranty for a period of 12 months from the date of Commissioning or 18 months from date of Dispatch whichever is earlier. |
| 9. | GST | As per the Commercial Table. |
| 10. | Remarks | . |
| 11. | Absstem GST Details | 06AAWCA9492P1ZF - ABSSTEM INDUSTRIAL SOLUTIONS PRIVATE LIMITED |

Hope you will find our offer in line with your requirement.

Should you require any further clarification/information, please feel free to contact us.

Thanking you, and assuring you of our best attention at all times.

Very Yours truly,

For Absstem Industrial Solutions Pvt. Ltd.

Arifa Zaidi

Assistant Manager Business Development

9218108184

www.absstem.com